ALAGAPPA UNIVERSITY JOURNAL OF BIOLOGICAL SCIENCES (AUJBS)

Vol 1 No 1 March 2017 ISSN:

Reach at

VI Floor Science Block
Alagappa University
Karaikudi 630 003 Tamil Nadu India
alubiojournal@gmail.com

Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Parasitic isopods of the family Cymothoidae from Indian fishes

S Ravichandran[1]* and **G Ramesh Kumar[2]**

[1]Centre of Advanced Study in Marine Biology,
Faculty of Marine Science, Annamalai University, Parangipettai, Tamil Nadu, India

[2]Unit of Toxicology, Department of Zoology,
School of Life Sciences, Bharathiar University, Coimbatore

Received: 15.01.2017 / Accepted: 23.02.2017
Published online: 25.03.2017

Abstract Parasitic infestation is a cosmopolitan phenomenon in almost all the habitats, especially among the aquatic biota. Isopod parasites have received considerable scientific attention because they cause series damage to fishery resources. The present paper was made to understand the nature of isopod parasites of marine fishes of Indian coastal waters. A number of 52 parasitic isopod species belonging to 23 genera were distributed from 60 species of fishes. Their distribution of existing parasite host systems in Indian coast was mostly on pelagic fishes. Among the fishes *Ilisha melastoma* was much infested the infestation rate was very high (30.5 %). The present paper provides a general introduction to the subject with special emphasis on types of parasitic isopods and their harmful effects, diversity and different microhabitat of Indian isopod parasites for the first time in cumulative manner.

Key Words parasitic isopods; Indian waters; diversity, microhabitat; harmful effects

Introduction

Parasites are invertebrate organisms; some are free-living and can become opportunistic parasites; others require hosts for their survival and reproduction, and these are referred to as obligate parasites. Both opportunistic and obligate parasites are found in fish hosts but most parasitic diseases in fish are generally caused by obligate parasites. Studying the diversity of parasites is particularly relevant because sympatric diversification may be important in some parasite taxa, and because of the opportunity for independent tests of evolutionary hypotheses in the many separate lineages in which parasitism evolved. Our incomplete knowledge of existing parasite species-the result of a range of phenomena that includes inadequate sampling effort or the lumping of different cryptic species under one name-is not always a major obstacle for the study of parasite diversity. Patterns in the diversity of parasites may be associated with either host or parasite characteristics. The distribution of parasite diversity among host taxa does not simply reflect the species diversity of the host taxa themselves; life history and ecological traits of hosts appear to play important roles.

Isopod parasites

Crustacea and fishes were thrown into ecological propinquity by the very fact their aquatic existence. Both in marine and freshwater environments fishes are, undoubtedly, among the best potential hosts for any prospective parasite. Among the crustaceans parasites isopods have received considerable scientific attention because they cause series damage to fishery resources. Their body form varies from an easily recognizable

*sravicas@gmail.com

isopod to a relatively amorphous sac recognized as an isopod only from the less modified male found within the folds. Isopods form an Order in the crustaceans. The name isopod refers to all legs being similar in size and shape.

Types of parasitic isopods

The Cymothoid isopods observed on the body, buccal cavity or gill cavity of the host (Trilles, 1969; Brusca, 1981; Ravichandran *et al*, 2001; 2011

Gnathiid isopods differ in morphology and behavior, when compared to Cymothoid isopods. They are parasitic only during their larval (Praniza) periods. Gnathiids can be found on the body and buccal cavity of fishes, but also on anemons and tunicates (Monod, 1931). Bopyrid isopods are morphologically different from other parasitic isopods, which can be found on shrimps and other free swimming decapods. Their final host, decapods, is preceded by copepods. Aegidae and Cymothoidae are temporary and obligatory parasites in fishes, sometimes with very low host specificity; *Nerocila acuminata* Schiödte & Meinert, 1884, for instance, has been observed on as many as 40 species of fishes- Eastern Tropical Pacific (families Engraulidae, Atherinidae, Serranidae, Mugilidae and Embiotocidae). Although numerous studies on isopod parasites in Indian waters but no cumulative report on this aspects hence the present paper will fulfill the lacunae.

Harmful effects of isopods

Many species of fishes are infected by cymothoids (Crustacea, Isopoda, Cymothoidae). They are blood-feeding; several species settle in the buccal cavity of fish, others live in the gill chamber or on the body surface including the fins. Their life cycle involves only one host (Holoxenic cycle). Isopods are associated with many species of commercially important fishes around the world and cause significant economic losses to fisheries by killing, stunting, or damaging these fishes. They can also kill or impair immature fishes so that they do not survive. Isopods serve as important food items for a variety of animals. They are commonly seen as parasites on teleosts in tropical and subtropical waters (Trilles, 1986; Trilles *et al*., 2013; Brusca, 1981; Bunkley-Williams and Williams 1998; Ravichandran et al., 2001; 2016a, b; Rameshkumar *et al*., 2012). Cymothoids harm the fish in several ways. Mancae feed voraciously and easily kill fry and fingerlings through the tissue damage they cause. Permanently attached adults stunt the growth of fish and retard or inhibit reproduction, probably because of the nutritional drain though more subtle mechanisms such as through hormonal changes have not been ruled out. Those in the gill chamber are usually associated with stunted gills, partly from pressure atrophy and partly from damage associated with feeding and attachment. They have also been frequently associated with anaemia.

Those in the mouth affect the development of oral structures and may completely replace the tongue, as in *Ceratothoa oestroides*. Menhaden infested by the buccal parasite, *Olencira praegustator*, school separately from uninfested fish of the same age. The tissue-inhabiting forms such as *Ourozeuktes* sp., which form a pouch from a depression in the skin, cause pressure atrophy of adjacent muscle and visceral organs. Though cymothoids penetrate the skin with their pereopods and mouthparts, and the tissue-inhabiting forms maintain a small opening to the outside, little secondary

infection occurs. Isopods harm the fish in three ways.

1. Isopods can harmful by its very presence, the hard body causing pressure atrophy of the soft tissues of the fish.
2. The process of attachment inevitably ends in mechanical damage to the host.
3. The parasite inflicts damage of varying degrees by feeding at the expense of the host. The damage can be either local or generalized, depending on the type of food taken by the crustacean.

The severity of infestation depending on the age and size of the fish and the age and size of the parasite, as well as on the site of the parasite activity. Sometimes two or all three types combine to produce the harmful effects observed in the fish (Kabata, 1970).

Distribution

There are around 52 species of isopods in Indian coast. A number of 52 species belonging to 23 genera were distributed from 60 species of fishes. Majority of isopods reported from fishes are ectoparasites. Their distribution of existing parasite host systems in Indian coast was mostly on pelagic fishes. When compare to shell fishes fin fishes were much infested by isopods. Among the fin fish's clupids fishes are much infected especially *Ilisha melastoma* fishes infestation rate was very high (30.5 %) and lowest infection rate was recorded in *Thryssa mystax* 1.2% this may be due to broad size of their opercular opening for entry and attachment of isopods..

Microhabitat selection of isopod parasites

Cymothoids are a family of ectoparasitic isopods found on body, fins or inside buccal or gill cavity. The distribution of isopod parasites were examined in three different microhabitats in host fishes viz, buccal, branchial and body surfaces. They exhibited host and site specific occurrence (Table. 1).

Branchial parasites

The gill parasites were found to attach either in the right or left gill chamber of host or both the gill chamber. The orientation of the branchial parasites varied in most of the hosts. *Agarna* sp, *A.brachysoma, A.tartoor, Alitropus typus, Elthusa samariscii, Joryma brachysoma, J.hilsae, J.tartoor, Indusa malayi, I. pustulosa, Irona far, Lironeca puhi, L.circularis, Livoneca engravlidis, Mothocya plagulophora* and *Ryukyua circularis* infest the host with its ventral region facing the inner surface of the opercula and dorsal region deeply plunged in the gill arches towards the lower half of the operculum (Fig.1 & 2).

Buccal parasites

The parasites found in the buccal cavities occupied the floor of the lower jaw of the host, and clung firmly to the tongue with their heads oriented towards to the opening of the mouth. The *Cymothoa* and *Glossobius* genera were found infesting the buccal cavity of the host fishes. *Codonophilus* sp, *Cymothoa cinerea, C.eremita, Cymothoa krishnai, C. indica, Cymothoa sp Joryma brachysoma, N.poruvae* were found in the buccal region of host fishes. *Joryma brachysoma* and *Nerocila*.sp were found attached bucco-pharyngeal region of their hosts *T.lepturus* and *Carangids malabaricus* respectively (Fig.3).

Body surface parasites

The isopod parasites were attached to the outer body surfaces of the fishes. *Anilocra leptosoma, A.longicauda, A.dimidiata, Livoneca* sp, *Nerocila* sp, *N.madrasensis, N. phaeopleura, N. pigmentata, N.poruvae, N.longispina, N. priacanthusi, N.pulicatensis, N.sundaica* and *N. trivitattata* found attached to the dorsal- lateral regions and near the pectoral fin. Isopods make frequent shifts in position on the host can cause a series of wounds. The parasites often move about as they feed. The digestive secretions of isopods apparently cause significant damage to the host's tissue (Fig.4).

Among the three microhabit (branchial, buccal, body surface) of isopods maximum site selection was recorded in the branchial region (Figure 3). The reason for maximum attachment is due to protective micro environment, especially heavy water currents and enemical waves, continuous water circulation as a system of ventilation, dispersal of their young once, facilitate easy shading and protected their eggs and young ones. More infection was recorded in Southwest coast of India compare to east coast of India.

Diversity of parasitic isopods

The order Isopoda contains about 9500 species that inhabit marine, fresh water and terrestrial ecosystems. According to Trilles (1986), 46 species of Cymothoidae have been reported in Africa (12 Anilocrinae and 34 Cymothoinae). Later himself (Trilles, 1991) listed 32 species cymothoa (Fabricius, 1793) from marine fishes world wide with 5 of these said to occur in Brazilian waters. The cymothoid fauna of diverse localities along the Indian coasts have been incompletely studied. Totally 52 species of cymothides were reported from various studies in both east and west coast of India. Though most adult isopods on fish belong to the Cymothoidae, there are parasitic forms in other families. The Aegidae, distinguished from Cymothoidae by having less modified pereopods, includes the notorious *Alitropus typus*, which parasites fishes in India and south-east Asia in fresh and brackish waters. The parasites attack fish to feed but retain their free-swimming capability as adults. Unlike cymothoids they do not appear to be protandrous hermaphrodites. Tridentellid and corallanid isopods are mostly free living but have a few representatives that are parasites of fish, such as the corallanid *Argathona macronema* which is common in the nasal passages of serranids and lutjanids on the Great Barrier Reef. Some corallanids are parasitic on Crustacea. Those belonging to the genus, *Tachaea*, are parasites of freshwater shrimps in Asia and Australia, and are usually found attached to the outside of the cephalothorax.

Conclusion

The ultimate aim of the study of the effects produced by the parasite on its host fish is usually assessment of the extent to which the harmful influence can affect the economic aims of man. The present paper was made to understand the nature of isopod parasites of marine fishes of Indian coastal waters. Fifty two species of parasites were recorded from 60 species of fishes. More infection was reported on fin fishes than shell fishes. Among the fin fishes pelgic fishes are more infested especially *Ilisha melastoma* fish. The loss of weight and retardation of the rate of growth under the influence of isopod have been reported. The parasites occupy the entire branchial chamber of the host may produce pressure on the gill surface and thus affecting

the efficiency of respiration. Although, the infestation may cause immediate death, it will affected the normal growth of the host fishes particularly degeneration of sexual organs. They may lead to economic loss of fishes. In general parasitic infection of fin and shell fishes mainly depend upon host factors such as age, size, sex, maturity, stage, behaviour, feeding and breeding, life cycle and particularly factors.

Acknowledgements

Authors are thankful to Department of Science and Technology (Ref. No.:- SB/SO/AS/063/ 2013) (DST), Government of India for providing financial support.

Fig. 1: *Trichurus lepturus* on *Nerocila sp*

Fig. 2: *Ilisha meslastoma* on *Joryma* *sp.*

Fig. 3a: Buccal parasites

Fig. 3b: Buccal parasites

Fig. 4a: Body surface parasites attached on *Leiognathus* sp and *Thryssa mystax*

Fig. 4b: Body surface parasites attached on *Leiognathus* sp and *Thryssa mystax*

Table 1

Branchial parasites	Buccal parasites	Body surface parasites
Agarna sp *Agarna brachysoma* *Agarna tartoor* *Alitropus typus* *Elthusa samariscii* ***Joryma brachysoma*** ***J.hilsae*** *J.tartoor* *Indusa malayi* *Indusa pustulosa* *Irona far* *Lironeca puhi* *L. circularis* *Livoneca engravlidis* *Mothocya plagulophora* *Ryukyua circularis*	*Codonophilus* sp *Cymothoa cinerea* *C.eremita* *Cymothoa krishnai* *C. indica* *Cymothoa* sp	*Anilocra leptosoma* *A.longicauda* *A.dimidiata* *Livoneca* sp *Nerocila* sp *N.madrasensis* *N. phaeopleura* *N. pigmentata* *N.poruvae* *N.longispina* *N. priacanthusi* *N.pulicatensis* *N.sundaica* *N.trivitata*

References

Brusca, R.C. (1981). A monograph on the Isopoda Cymothoidae (Crustacea) of the Eastern Pacific. *Zoological Journal of the Linnean Society*, 73 (2): 117-199..

Bunkley-Williams, L. & E.H. Williams, Jr. 1998. Isopods associated with fishes: A synopsis and corrections. *J. Parasit*. 84: 893-896.

Kabata, Z. 1970. Crustacea as enemies of fishes. -In: S. F. Snieszko and H. R. Axelrod (eds.), Diseases of fishes, Book 1. T.F.H. Publishers, Jersey City, New Jersey. Pp. 1-171.

Monod T, 1931. Crustacés de Syrie. In: *Les états de Syrie.Richesses marines et fluviales. Exploitation actuelle et avenir* (ed. A. Gruvel), Bibliothèque de la faune des colonies françaises, société d'editions géographiques, maritimes et coloniales, France, pp. 397-435

Rameshkumar, G., S. Ravichandran., and J.P.Trilles 2012. Observation on an isopod parasitizing the edible fish *Parastromateus niger* in the Parangipettai coast of India. *Journal of Environmental Biology*, 33:191-193

Ravichandran, S., A.J.A. Ranjith Singh and N. Veerappan, 2001. Parasite induced vibriosis in *Chirocentrus dorab* off Parangipettai coastal waters. *Current Science*, 80: 101-102.

Ravichandran.S., G. Rameshkumar, and J.P.Trilles. 2011. New records of two parasitic cymothoids from Indian fishes. *Journal of Parasitic Diseases*, 35(2): 232-234

Ravichandran **S,**K. Sivasubramanian, P Parasuraman, D. Karthick Rajan and G. Ramesh kumar.2016a. Isopod Parasite Induced Secondary Microbial Infection in Marine Food Fishes. *J. Fish Pathol*., 29(1) : 1~5

Ravichandran, S., Kanagasabapathy Sivasubramanian, Ganapathy Rameshkumar, and Narayanasamy Veerappan. 2016b. High prevalence and infestation of *Mothocya renardi* (Isopoda, Cymothoidae) in marine fish *Strongylura leiura* (Bleeker 1850). J Parasit Dis DOI 10.1007/s12639-015-0696-7.

Schioedte JC, Meinert F.1884. Symbolae ad monographiam Cymothoarum Crustaceorum Isopodum Familiae. IV. Cymothoidae. TRIB. II. Cymothoinae. TRIB. III. Livonecinae. Naturhistorisk Tidsskrift, Ser. III, XIV: 221–454, pls VI. XVIII (Cym. XXIV.XXXVI).

Trilles JP, 1969. Recherches sur les Isopodes Cymothoidae des côtes Françaises. Aperçu général et comparatif sur la bionomie et la sexualité de ces crustacés, *Bull Soc Zool*, Fr., 94(3): 433- 445.

Trilles JP (1986). Les Cymothoidae (Crustacea, Isopoda, Flabellifera) d'Afrique. Bull. Mus. natn. Hist. nat., Paris, 4e sér., 8, section A. n° 3 : 617-636.

Trilles, J.P. 1991. Les Cymothoidae (Crustacea, Isopoda) du monde. Studia Marina 21/22 (1-2): 5-288.

Trilles J.P., G. Rameshkumar and S. Ravichandran 2013. *Nerocila* species (Crustacea, Isopoda, Cymothoidae) from Indian marine fishes. Parasitol Research. 112(3): 1273-1286.